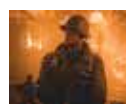




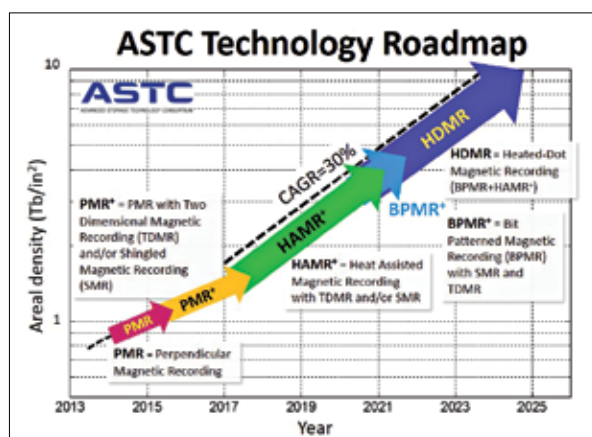
Facebook to design chips

Facebook is in the early stages of building a team, in an attempt at self-reliance, to develop their own chips. <https://dgit.in/fbchips>



New COD WWII Coach: Alexa

Alexa now offers coaching in Call of Duty: WWII based on AI and machine learning that analyses how you play. <https://dgit.in/alexacod>



Storage capacity has grown by leaps and bounds over the years

and that has enabled hard drives of 8 TB and above possible. However, your next hard drive is likely to use something called HAMR (Heat Assisted Magnetic Recording).

HAMR is quite promising because it has the potential to change the rate of growth of areal density by 30 percent per year. That's double the current trend of 15 percent. If the new nanoparticle-based film storage technology enters mass production, then we're looking at the largest spike in areal density that the storage industry will have seen.

MADE IN CHINA

Currently under development at the Northeast Normal University, China, this new technology is a different storage medium altogether. So no, your hard drives are not going to be enhanced by this technology. It's being built to record holographic information. The nanoparticle film in question can be incorporated into a tiny storage chip that records 3D colour information which can be realised as a 3D hologram. That too, with realistic detail.

Moreover, it's quite durable as well. Shencheng Fu, the lead researcher on the project claims that the storage medium is environmentally stable to the extent that it can function in the harsh radiation conditions of outer space. They not only hold massive amounts of data but the same data can be retrieved at speeds of up to 1 Gbps.

they change colour when hit by different wavelengths of laser light. Also, the method can be used to store a set of 3D images using laser in a single step. However, there is a drawback to this technology, one that has plagued silicon for quite some time – Ultra-violet Light. Exposure to UV light can erase the data stored on these medium, hence, it becomes unviable for long-term data storage. You can try this out yourself by taking a UV LED and flashing it over the window atop an EPROM chip.

This vulnerability exists because recording a holographic image onto titania-silver medium requires the use of a laser that converts the silver particles into silver cations because it gained extra electrons. Unfortunately, UV light has the same effect as the laser on the medium. When struck by UV light, the electrons from the semiconductor film transfer onto the silver nanoparticles, and thus converts the silver particles into cations. This results in the data getting erased. The



Shencheng Fu, the lead researcher on the project.

NANOPARTICLE FILMS

In order to understand the new medium, you need to know the drawbacks of similar medium that already exist. As of now, porous films made out of semiconductor titania and silver nanoparticles have shown much promise for storing nanoscale holograms because

key change in the new research by Shencheng Fu was the introduction of electron-accepting molecules into the medium. When struck by UV light, the excess electrons would be drawn to the new molecules rather than the silver particles. This mitigated the effect of UV light and made the medium more stable and durable for use in the hazardous radiation conditions of space.

In order to make the new films, researchers had to use electron-accepting molecules that were only 1-2 nanometers thick. They fabricated the nanoparticle films with a honeycomb nanopore structure so that all three components, i.e. the silver nanoparticles, the semiconductor and the electron-accepting molecule could all



One zap and it's all gone!

interface with each other. As a result, the final nanoparticle films were just 620 nm thick.

SO WHEN CAN I BUY IT?

Not anytime soon. While the technology has been proven to be durable and capable of storing massive amounts of data, it will remain within the confines of the research facility for the time being. That's because the researchers indicate that real-world application of these nanoparticle films would require the development of high efficiency 3D image reconstruction techniques and also methods to depict data that's stored as different colours. Until these things are developed and miniaturised to fit in your average hard drive, you'll have to stick with HAMR hard drives or maybe the MAMR (Microwave Assisted Magnetic Recording) drives which are right around the corner.